

8.8 Classwork: Transformations and Composition

Part A — The language of transformations

A *full* description names three things: the **type**, the **direction**, and the **amount**.

Word bank: translate (shift, slide) • stretch (dilate, scale) • reflect • vertical • horizontal • scale factor • up • down • left • right

A1. Fill in each blank with a word from the bank.

- (a) A _____ slides every point the same distance in the same direction.
- (b) To describe a horizontal translation fully, give the number of units *and* state _____ or _____.
- (c) A vertical _____ with _____ 2 doubles every y -coordinate.
- (d) Multiplying every output by a number is also called a _____ (another word for stretch).
- (e) In $y = \frac{a}{x} + k$, the $+k$ produces a vertical _____.
- (f) The vague word “_____” does *not* count as a description — use translate, shift, or slide instead.

A2. The graph of $y = \frac{1}{x}$ is changed to $y = \frac{1}{x} + 4$. Circle every description that earns full credit; cross out those that do not.

- (a) move up 4
- (b) translate 4 up
- (c) shift 4 in the positive y -direction
- (d) $+4$
- (e) slide vertically 4

A full description needs a proper word and (for horizontal moves) an explicit direction; bare symbols do not count.

A3. Draw a line matching each term to an equivalent term.

translate	dilate
stretch	flip
reflect	slide

Part B — Describe the transformations

For each function the parent graph is $y = \frac{1}{x}$. **Describe fully, in words**, the sequence of transformations, and write the equations of the two asymptotes.

1. $f(x) = \frac{1}{x-2} + 3$

2. $f(x) = \frac{3}{x} - 1$

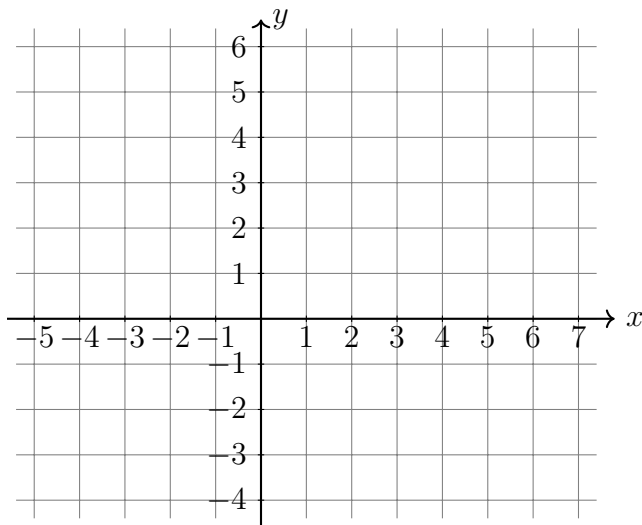
3. $f(x) = -\frac{1}{x}$ *(this one includes a reflection)*

4. $f(x) = \frac{2}{x+4}$

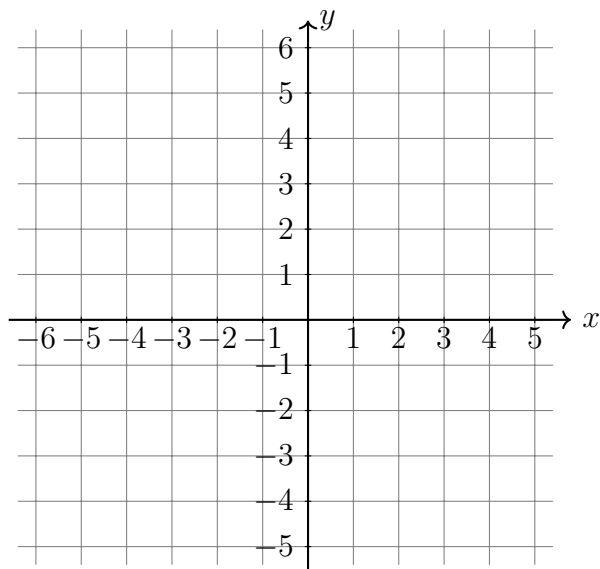
Part C — Sketch the transformed graph

Sketch each graph on its grid. Draw the asymptotes (or axis of symmetry) as **dashed** lines and label them, then plot at least two points.

1. $y = \frac{1}{x-2} + 1$



2. $y = -(x+1)^2 + 4$ (a transformed parabola)



Part D — The placeholder idea

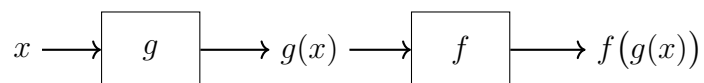
The x in a formula is just a **placeholder** for whatever you feed in. If $f(x) = 3x - 4$, then $f(\text{anything}) = 3(\text{anything}) - 4$.

D1. Let $f(x) = 3x - 4$. Fill in each output.

$$f(7) = \underline{\hspace{2cm}} \qquad f(-2) = \underline{\hspace{2cm}} \qquad f(a) = \underline{\hspace{2cm}}$$

$$f(2x) = \underline{\hspace{2cm}} \qquad f(x+1) = \underline{\hspace{2cm}} \qquad f(g(x)) = \underline{\hspace{2cm}}$$

D2. A composition is a chain of machines. The output of one becomes the input of the next.



Which machine acts *first*, f or g ? $\underline{\hspace{2cm}}$

D3. Let $f(x) = 3x - 4$ and $g(x) = 2x + 2$.

(a) Find $f(g(x))$ by feeding $g(x)$ into f .

(b) Find $g(f(x))$.

(c) Are they equal? What does this tell you about the order of composition?

Part E — Composition and recovering a hidden function

Read $f \circ g$ as “ g goes first, then f wraps around it”: $(f \circ g)(x) = f(g(x))$. $\circ$ is **not** multiplication.

E1. The function f is defined by $f(x) = 3x - 4$, and $(f \circ g)(x) = 6x + 2$. Find $g(x)$.

The input to f is now $g(x)$, so $f(g(x)) = 3g(x) - 4$. Set this equal to $6x + 2$ and solve for $g(x)$.

E2. The function f is defined by $f(x) = 2x + 1$, and $(f \circ g)(x) = 2x + 7$. Find $g(x)$.

E3. Recover the outer function. Given $g(x) = x - 3$ and $(f \circ g)(x) = 4x - 5$, find $f(x)$.

Hint: let $u = x - 3$, so $x = u + 3$; write $f(u)$ in terms of u .

E4. Spot the error. A student claims $g(x) = \frac{6x + 2}{3x - 4}$ in E1. In one sentence, explain why this is wrong.

Part F — The inverse view, and always check

Another way to recover g : if f acts on $g(x)$, then f^{-1} *undoes* f and hands back $g(x)$: $g(x) = f^{-1}((f \circ g)(x))$.

F1. For $f(x) = 3x - 4$ (from E1):

(a) Find $f^{-1}(x)$.

(b) Compute $g(x) = f^{-1}(6x + 2)$ and confirm it matches your answer to E1.

F2. The check habit. Verify E1 by substituting your g back: compute $f(g(x))$ and confirm it equals $6x + 2$.

F3. Why the inverse view matters. When the outer function is *not* linear, substitution is awkward but the inverse still works. Given $f(x) = x^3$ and $(f \circ g)(x) = 8x^3$, find $g(x)$.

Hint: $f^{-1}(x) = \sqrt[3]{x}$.